ADRC hands-on workshop for single-nuclei RNA-seq (Week 1)

Kevin Z. Lin and Katherine E. Prater

July 27, 2026

Introduction to us

- Kevin Z. Lin (kzlin@uw.edu) : Biostatistician, 8 years experience in snRNAseq analysis and developing new statistical/computational methods for it

- Katherine E. Prater (keprater@uw.edu) : Neuroscientist/biologist by training, 6 years experience in snRNAseq analysis

Introductions

- Let’s go around and briefly state:

    1. Your name (pronouns)
    2. Your lab
    3. Your degree/program/role
    4. What you are scientifically studying overall
    5. A bit about the dataset you’ll be working with during this workshop
    6. What related to aspects of single-cell RNA-sequencing analyses you’ve had exposure to

Logistics

- KZL isn’t able to get you building access, so each week, you’ll be picked up from the 1st floor lobby by either me or Katie

- Feel free to directly text me (848-466-6541) or KEP (425-269-9251) or email me
    - Send KZL a text with your name so he can not accidentally ignore your texts/call

- We’ll record about a 30-minute presentation at the start of each week, followed by a <2 hr hands-on session, concluded by a 15 minute discussion/recap. (2:30-5pm)

- We’ll sprinkle throughout the workshop how KZL uses Claude Code in his analyses workflow.

Resources for this workshop

- GitHub: https://github.com/linnykos/ADRC_workshop_2026
    - This is where the code we showcase will be deposited
    - Has the Google Slide link to these slides

- BIOST 545 lecture notes: https://linnykos.github.io/scOmicNotes/
    - Lecture notes written by Kevin that go more into the conceptual understanding of single-cell methods and its usage in different omics

The recorded content for Week 1

Intent and goals of this ADRC workshop

- snRNA-seq requires quite a complex set of computational skills to analyze

- There is no “one-and-done” workflow that always will work, so it’s behooving to learn how each component “works” at a high level so you can make educated decisions on how to steer your analysis

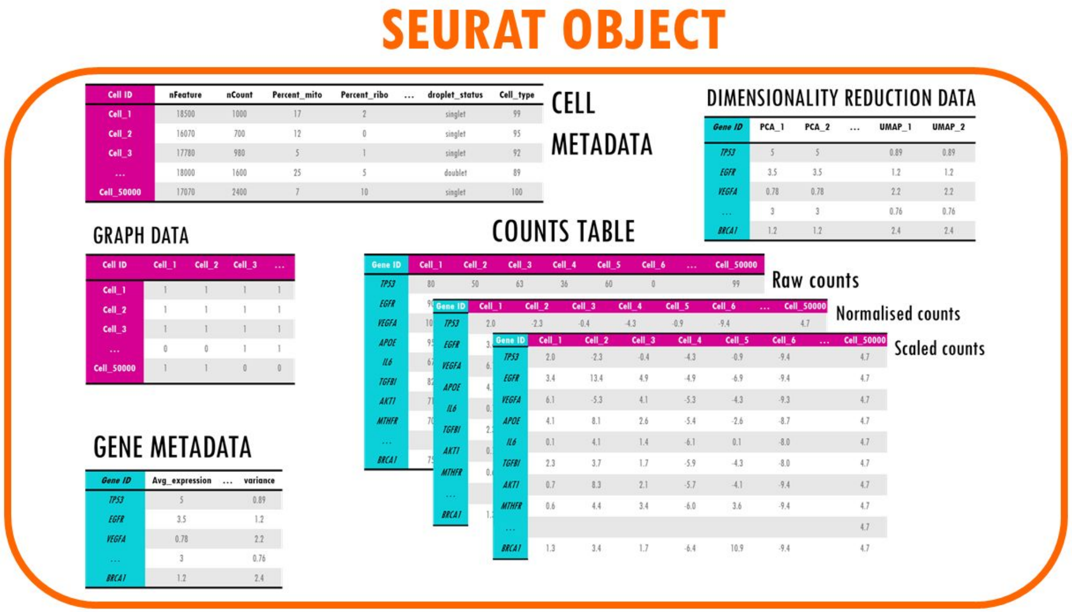
- There is no “optimal” analysis per-se, but there are certainly analyses that are less justifiable than others.

The bird’s-eye view of this workshop

- **Week 1:** The high-level workflow
- **Week 2:** Ambient RNA removal, doublet detection, and discussion on sequence alignment
- **Week 3:** Normalizing gene expression counts, selecting genes to analyze, clustering
- **Week 4:** Differential expression, cell-type labeling
- **Week 5:** Batch correction, bulk deconvolution

What is a “Seurat object”?

- See: https://biostatsquid.com/seurat-objects-explained/
- (Also: https://rnabio.org/module-08-scrna/0008/01/02/scRNA_Data/ and https://x.com/lpachter/status/1524413513233575936)



Basics of Seurat

- How to access the cell-level "metadata": `pbmc@meta.data`

```
> pbmc@meta.data
                   orig.ident nCount_RNA nFeature_RNA
AAACATACAACCAC-1      pbmc3k       2421          781
AAACATTGAGCTAC-1      pbmc3k       4903         1352
AAACATTGATCAGC-1      pbmc3k       3149         1131
AAACCGTGCTTCCG-1      pbmc3k       2639          960
AAACCGTGTATGCG-1      pbmc3k        981          522
AAACGCACTGGTAC-1      pbmc3k       2164          782
```

- How to access the underlying count data: `SeuratObject::LayerData(pbmc, layer = "counts", assay = "RNA")`

```
> count_mat <- SeuratObject::LayerData(pbmc,
+                                      layer = "counts",
+                                      assay = "RNA")
> count_mat[42:44,14:16]
3 x 3 sparse Matrix of class "dgCMatrix"
             AAAGAGACGCGAGA-1 AAAGAGACGGACTT-1
RP11-5407.11                .                .
ISG15                       5                .
AGRN                        .                .
             AAAGAGACGGCATT-1
RP11-5407.11                .
ISG15                       2
AGRN                        .
```

Basics of Seurat

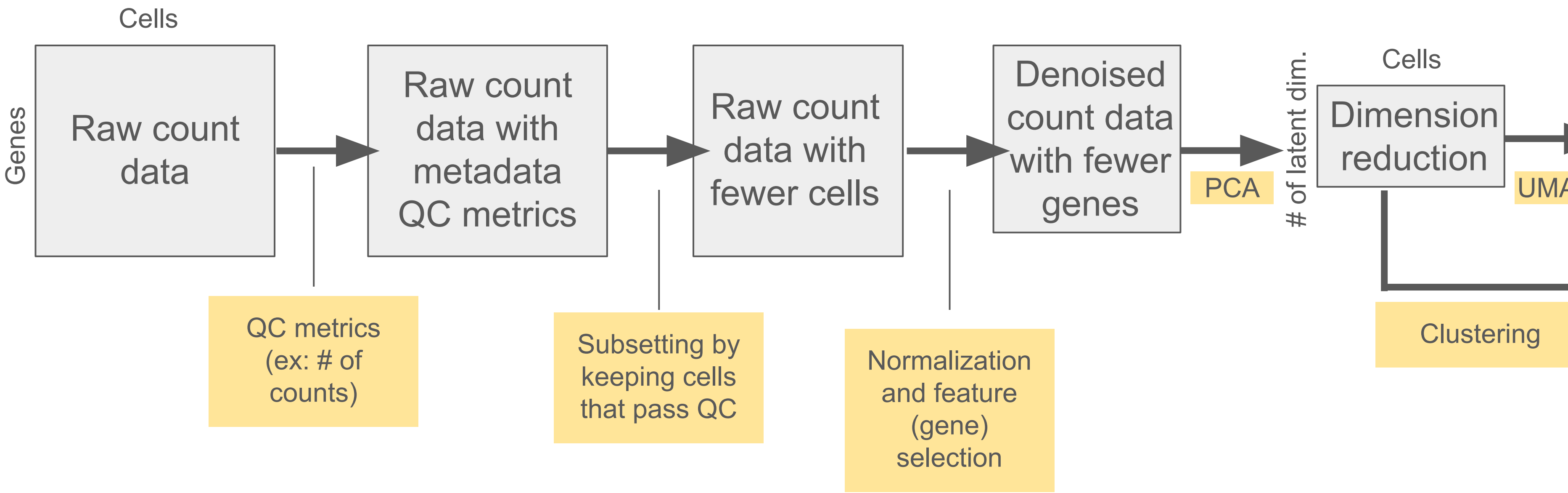
- How to access the genes in the dataset:
  `SeuratObject::Features(pbmc)`

```
> SeuratObject::Features(pbmc)[1:5]
[1] "MIR1302-10"    "FAM138A"       "OR4F5"
[4] "RP11-34P13.7" "RP11-34P13.8"
```

- How to access the cell barcodes in the dataset: `Seurat::Cells(pbmc)`

```
> Seurat::Cells(pbmc)[1:5]
[1] "AAACATACAACCAC-1" "AAACATTGAGCTAC-1"
[3] "AAACATTGATCAGC-1" "AAACCGTGCTTCCG-1"
[5] "AAACCGTGTATGCG-1"
```

The "typical workflow" (for today)



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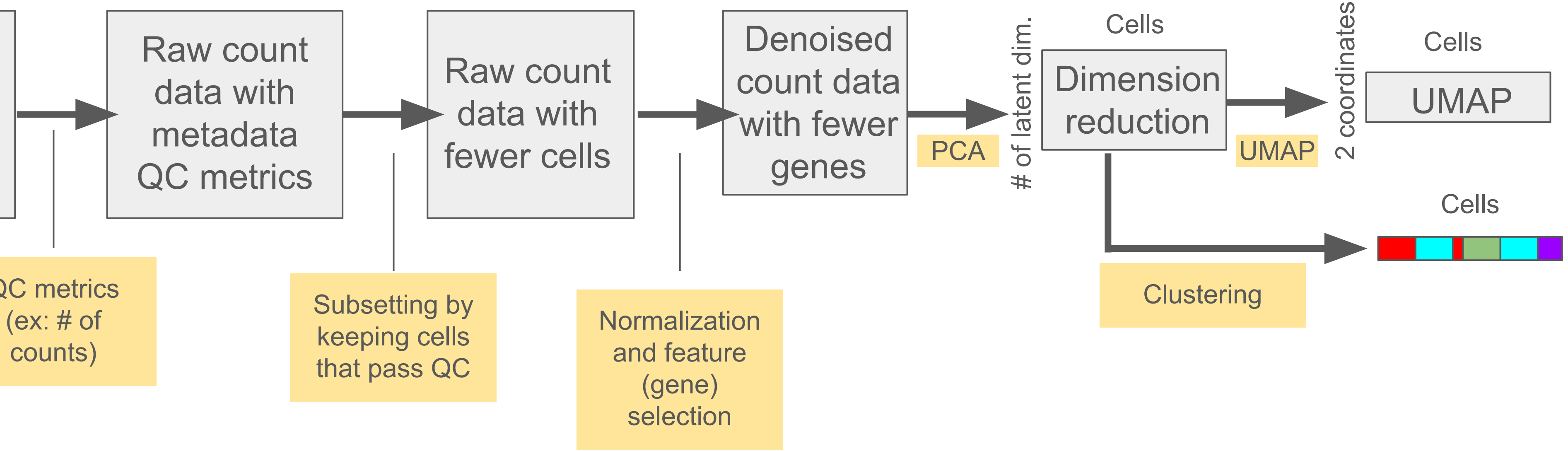
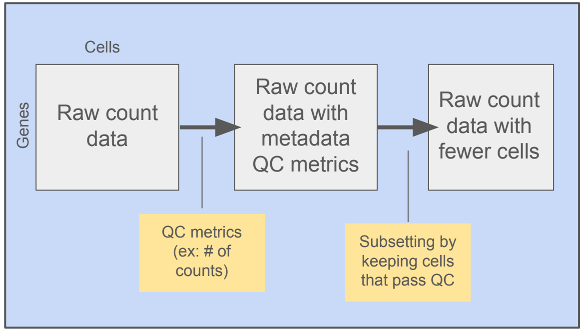
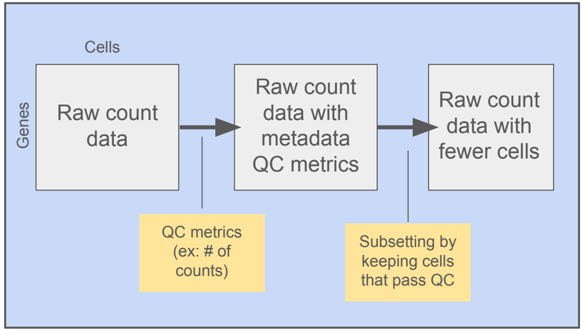


Step 1: Quality control and cell filtering



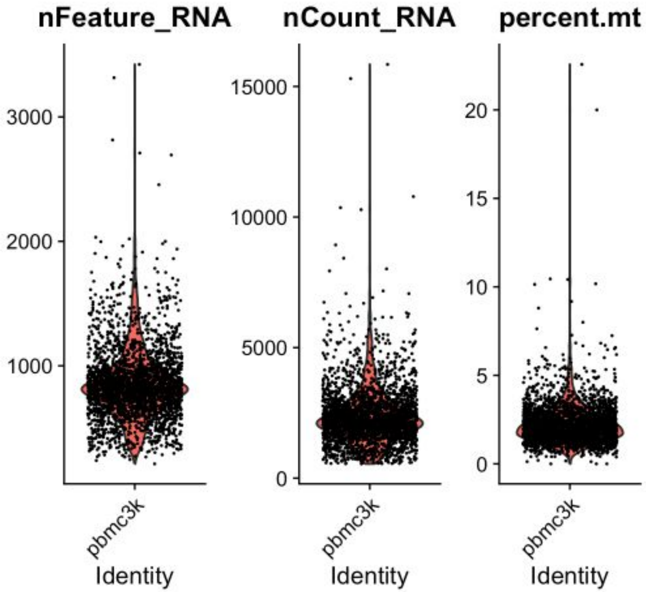
Can we filter out some “poor quality” cells beforehand?

Step 1: Quality control and cell filtering

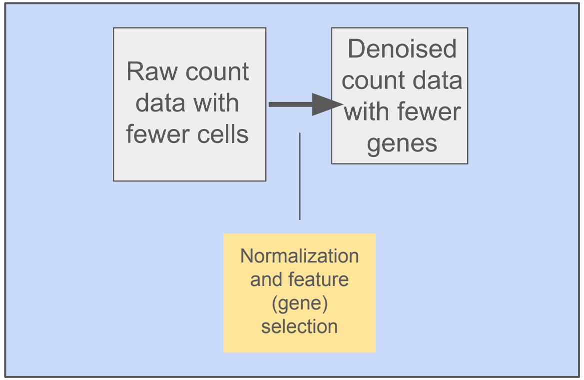


Can we filter out some “poor quality” cells beforehand?

- `gene_vec <- SeuratObject::Features(pbmc)`
- `gene_vec[grep("^MT-", gene_vec)]`
- `pbmc[["percent.mt"]] <- Seurat::PercentageFeatureSet(pbmc, pattern = "^MT-")`
-
- `## Visualizing the QC metrics`
- `Seurat::VlnPlot(pbmc, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)`



Step 2: Normalizing the cell's gene expression



Since single-cell sequence experiments are literally *counting* the # of reads that map to a genomic region, often times, the literal count itself is not informative. Instead, the *relative* count is.

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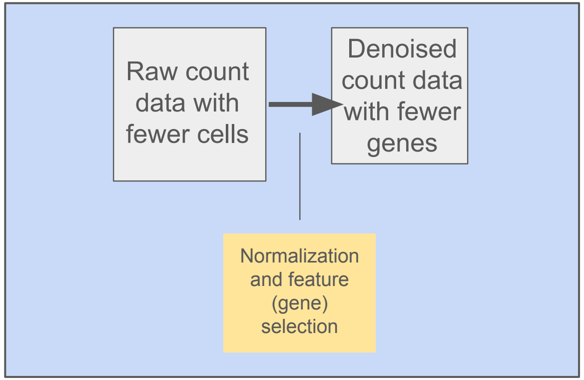

Cells in condition 1 | Cells in condition 2

Gene: 1, 1, 2, 0, 1, 2, 1, 2, 0, 0, **23, 10, 12, 20, 10, 15**

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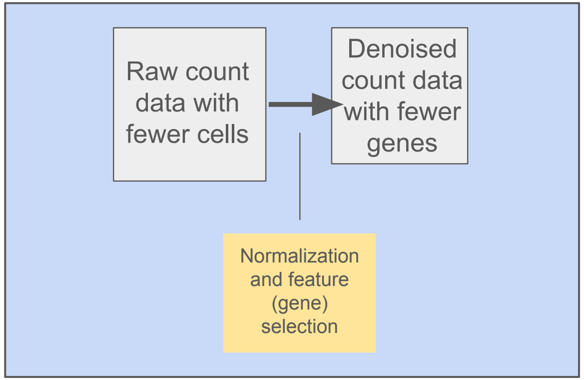
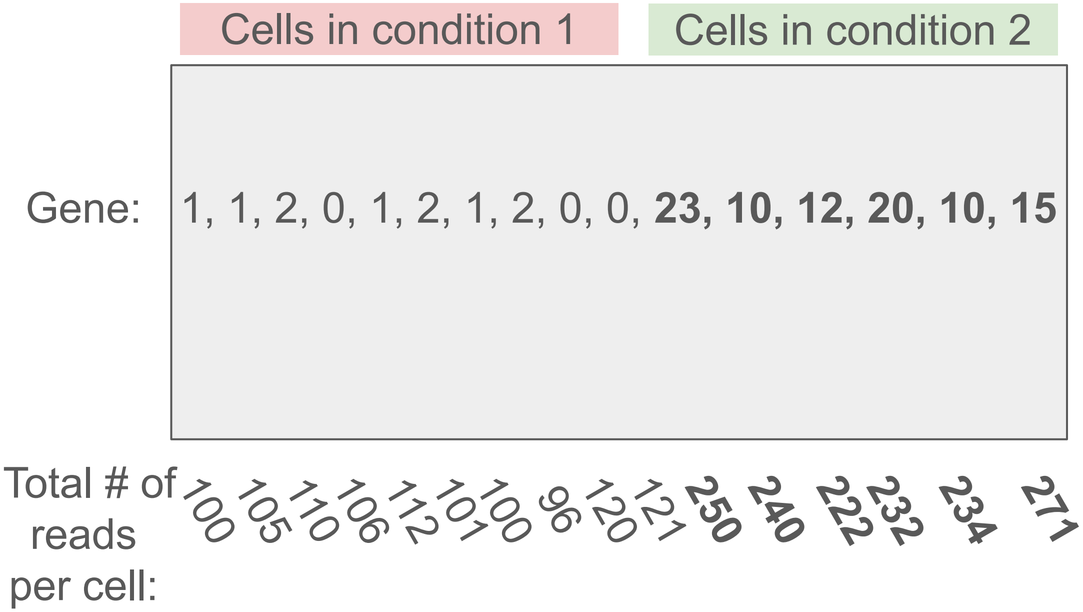
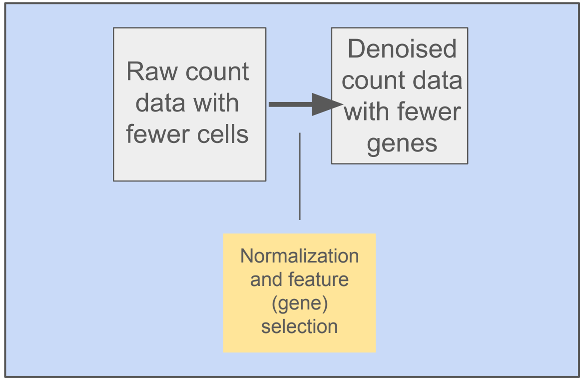


Since single-cell sequence experiments are literally *counting* the # of reads that map to a genomic region, often times, the literal count itself is not informative. Instead, the *relative* count is.

- `pbmc <- Seurat::NormalizeData(pbmc)`



Step 3: Filtering out highly variable genes



Computational biology is often subtractive – signal is so weak that if you try analyzing *everything*, you won't find much. Instead, focus on genes that likely show promise. Typically, this means focusing on genes with a good range of values (i.e., "highly variable")

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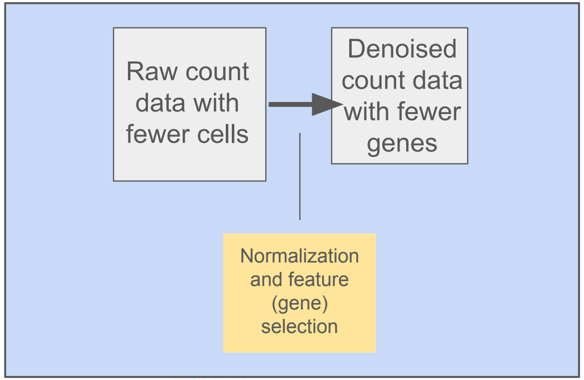
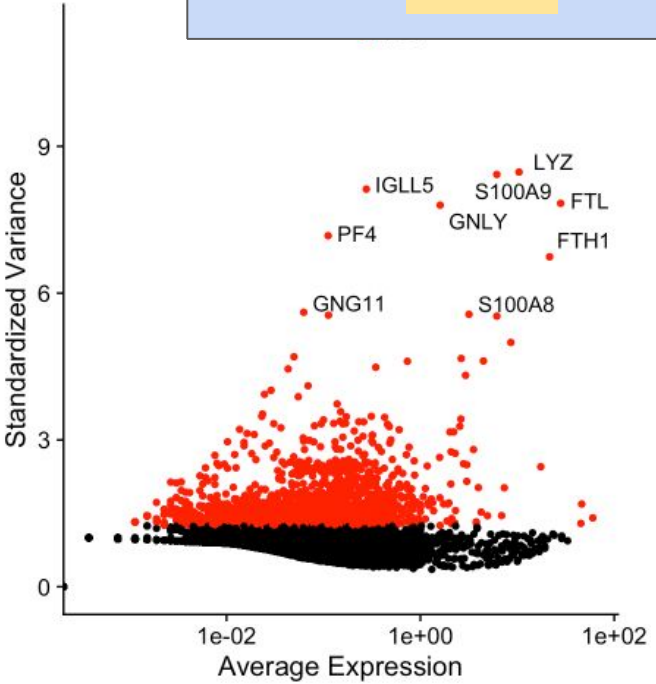
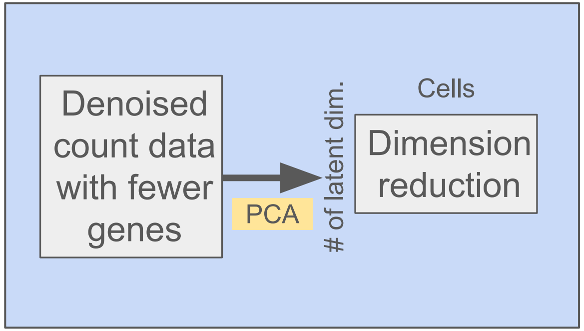
Computational biology is often subtractive – signal is so weak that if you try analyzing *everything*, you won't find much. Instead, focus on genes that likely show promise. Typically, this means focusing on genes with a good range of values (i.e., "highly variable")

- ```
  pbmc <- Seurat::FindVariableFeatures(pbmc, selection.method = "vst", nfeatures = 2000)
  ```



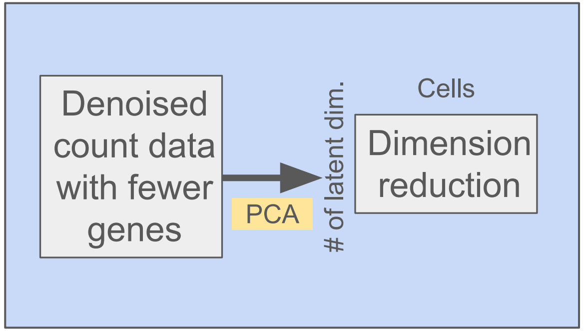

  ```

Step 4: Applying dimension reduction



An important step is to create a “low-dimensional” representation. Think of this as a concise summary for each cell, where each number represents a potential “direction” in gene-expression-space that unveils meaningful variation across all your cells.

Step 4: Applying dimension reduction



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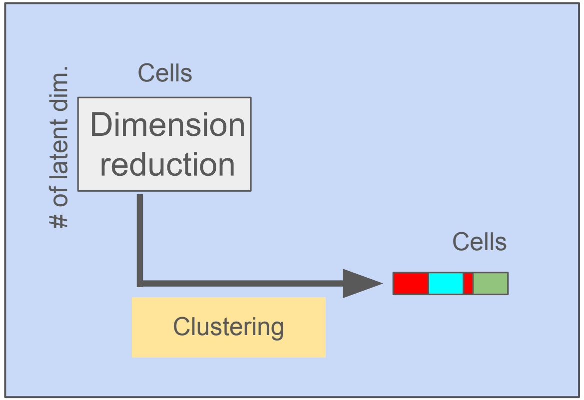
```
## Scaling
pbmc <- Seurat::ScaleData(pbmc)

## PCA
pbmc <- Seurat::RunPCA(pbmc,
                       features = Seurat::VariableFeatures(object = pbmc), verbose = FALSE)
```

Step 5: Clustering the cells

Often when working with a “fresh” dataset, you won’t have *any* labels on your cells. You need to make the labels yourself. One important intermediary step is to simply group your cells into “similar-ish gene expression profiles.”



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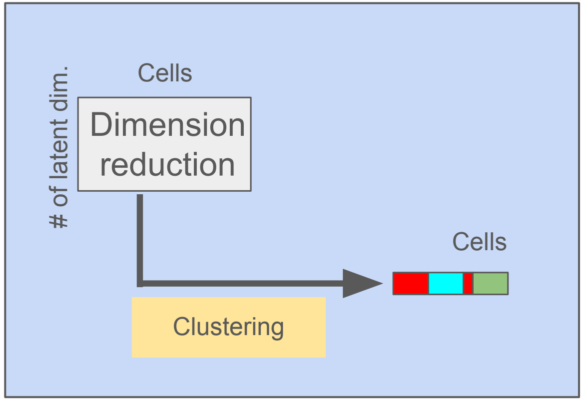

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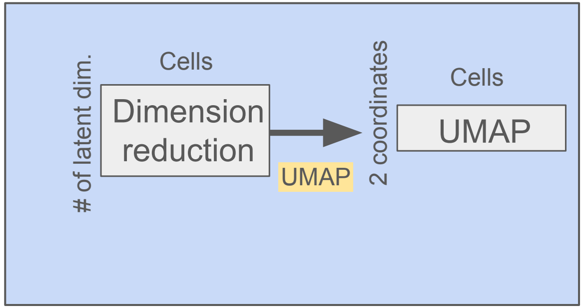
- `pbmc <- Seurat::FindNeighbors(pbmc, dims = 1:10)`
- `pbmc <- Seurat::FindClusters(pbmc,resolution = 0.5)`

Step 6: Visualizing the data



Now for the payoff – you're ready to visualize your data on a 2D plane. This is often called a "UMAP," and it's a plot to get a good qualitative summary of your data.

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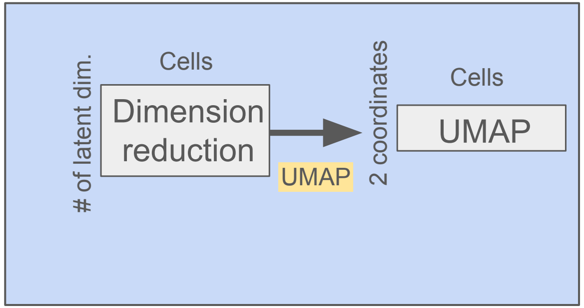
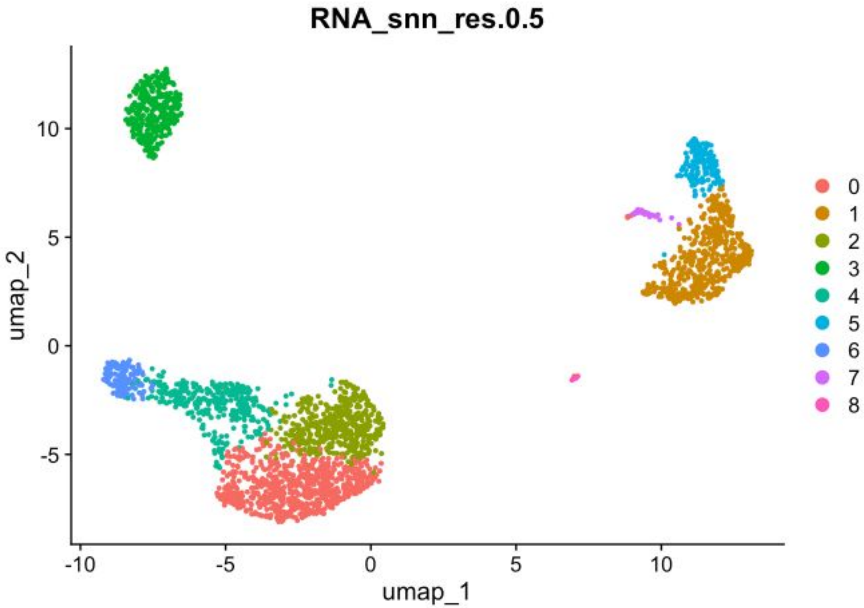
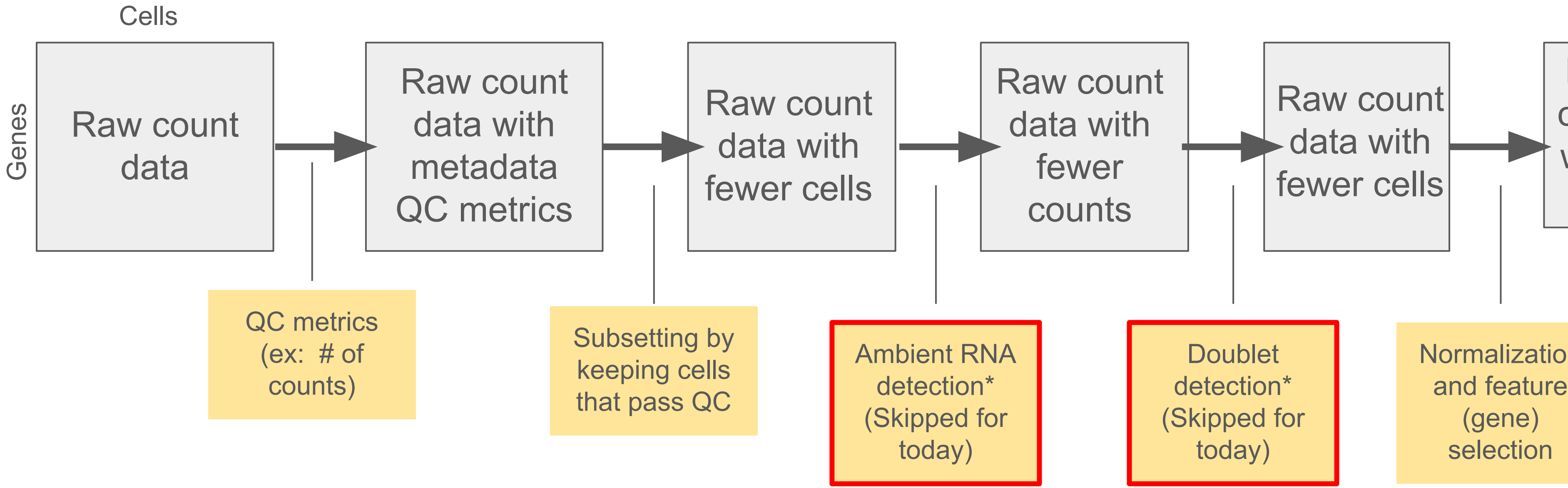


Now for the payoff – you're ready to visualize your data on a 2D plane. This is often called a "UMAP," and it's a plot to get a good qualitative summary of your data.

- `pbmc <- Seurat::RunUMAP(pbmc, dims = 1:10)`
- `Seurat::DimPlot(pbmc, reduction = "umap", group.by = "RNA_snn_res.0.5")`



How we're going to be adjusting the workflow in the coming weeks



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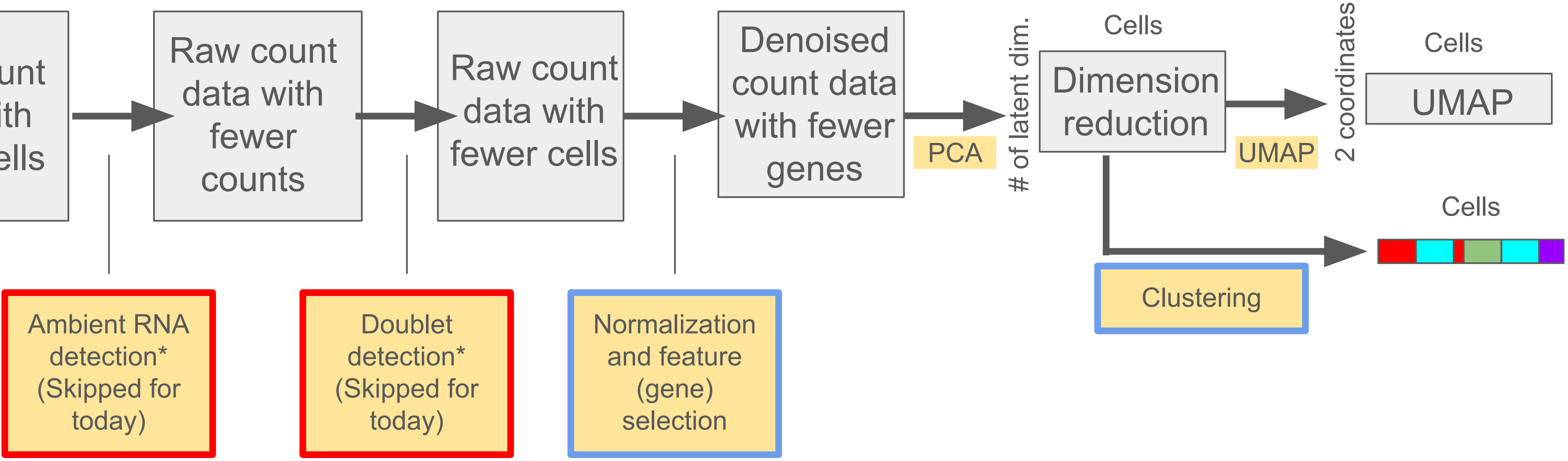

Appendix:

- See a useful cheatsheet: https://satijalab.org/seurat/articles/seurat5_essential_commands

- For a tutorial: You can follow https://satijalab.org/seurat/articles/pbmc3k_tutorial as our tutorial (see https://github.com/linnykos/ADRC_workshop_2026/blob/main/Week1_birdseye/pbmc_tutorial.R and the HTML file https://github.com/linnykos/ADRC_workshop_2026/blob/main/Week1_birdseye/pbmc_tutorial.html)
  - Download the data needed at https://www.dropbox.com/scl/fi/9mn6q4clnjobfdsmpgshs/seurat-pbmc.zip?rlkey=z4ha1mdwcsoew6o712ciwfq5i&dl=0

Start of the hands-on portion

What you can work on:

Generally, you can apply the tutorial in https://satijalab.org/seurat/articles/pbmc3k_tutorial but on your dataset.

- **Chris**: You can ignore the spatial component of your data for now. You can see https://github.com/linnykos/ADRC_workshop_2026/blob/main/Week1_birdseye/spe_to_seurat.R and https://github.com/linnykos/ADRC_workshop_2026/blob/main/Week1_birdseye/spe_workflow.R
- **Colin**: This pipeline should work smoothly for the ferret data I found.
- **Gwen**: Hopefully this pipeline works?
- **Ives**: Katie (Prater)'s data has already been processed, but you can redo this pipeline to build familiarity. We can discuss the setup on Hyak.
- **Karinn and Kathleen**: This might be a bit tricky with only 10 neurons, but on face value, it should work
- **Katie (Blakeman)**: Hopefully this pipeline works?
- **Xinmei**: This pipeline should work smoothly for just one subject's worth of data. We can discuss the setup on Hyak.